HOSPITAL EMERGENCY ROOM PERFORMANCE DASHBOARD

Project Report

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ABSTRACT

Modern healthcare environments operate under volatile, uncertain, complex, and ambiguous conditions, with the Emergency Room (ER) functioning as the primary gateway for critical patient care. The operational paradigm of an ER is characterized by severe resource constraints, variable patient inflows, and the non-negotiable requirement of minimizing clinical delay to prevent adverse clinical outcomes. This internship project introduces a full-scale clinical intelligence and resource allocation framework titled the "Hospital Emergency Room Performance Dashboard," built using Microsoft Power BI Desktop Version 2.153.1206.0 (64-bit) released in April 2026. The objective of this engine is to ingest, transform, clean, model, and visualize historical operational logs containing 9,216 distinct patient transaction records, thereby deriving actionable intelligence that can optimize clinical operations, reduce patient wait times, and improve healthcare delivery structures.

The structural core of the project relies on a highly optimized Star Schema Data Warehouse Model. The data architecture isolates operational transactions in a centralized fact table (Hospital ER_Data) and relates them to an engineered, highly granular temporal dimension table (Date Table). This dimensional framework facilitates time-intelligence operations using advanced Data Analysis Expressions (DAX). To evaluate operational efficiency and clinical bottlenecks, a matrix of customized Calculated Columns and structural Measures was developed. Key calculations include specific patient full-name concatenations, absolute date parsing, operational wait-time threshold flagging (Within Target vs. Target Missed based on a strict 30-minute operational SLA), and chronological hour extraction. Advanced DAX formulations were developed to compute intricate clinical operational parameters, such as the Dynamic Peak Day, Dynamic Peak Hour, Net Patient Inflow, Satisfaction Score, and Time Average. These measures allow hospital leadership to track real-time resource utilization, administrative backlogs, and patient-reported clinical experiences.

The analytical interface is segmented across six distinct interactive dashboard layers: (1) Monthly View, provides an overarching strategic view of volumetric variations and key performance indicators (KPIs) over a temporal horizon; (2) Consolidated View, aggregates executive-level operational metrics to provide a macro-level view of institutional health; (3) Patient Details, offers micro-level descriptive analytics for granular line-item case audits; (4) Key Takeaways, translates quantitative data distributions into direct, high-level structural advice for hospital boards; (5) Demographics & Diversity, breaks down clinical demand patterns across age bands, gender variations, and racial groups; and (6) Operations & Peak Hours, evaluates chronological bottlenecks, isolating specific hourly blocks where incoming patient volume exceeds standard staffing capacities.

The business intelligence derived from simulating specific operational scenarios reveals that administrative backlogs are heavily concentrated during specific peak hour intervals (13:00 to 16:00 and 19:00 to 22:00) on Mondays and Tuesdays. By implementing the predictive insights, operational managers can execute proactive staffing realignments, dynamic bed-allocation algorithms, and predictive triage routing. The end-to-end framework bridges the gap between raw unstructured clinical logs and executive strategic planning, showcasing how automated business intelligence solutions built with Power BI can drive down operational costs, ensure compliance with clinical service level agreements, and directly improve patient satisfaction and outcomes.

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CHAPTER 1: INTRODUCTION

The deployment of Business Intelligence (BI) in modern hospital administration has shifted from an operational luxury to a core requirement for survival. Healthcare providers globally operate within high pressure environments where resource constraints intersect with critical, time-sensitive patient needs. Among all hospital segments, the Emergency Room (ER) stands out as the most complex, volatile, and logistically challenging unit. Managing an ER requires continuous balancing of patient inflows, severity of illnesses, clinical staging, staff rostering, and diagnostic turnaround times. This project addresses these administrative challenges by designing and deploying an interactive data analytics engine: the Hospital Emergency Room Performance Dashboard, implemented via Microsoft Power BI.

1.1 What is Power BI

Microsoft Power BI is an advanced cloud-enabled business intelligence and data visualization architecture that translates raw, disconnected streams of transactional records into structured visual analytics and predictive data models. The tool bridges the gap between complex backend databases and business managers. By utilizing a hybrid computational model combining the VertiPaq high-performance in-memory column-store database engine with data transformation interfaces (Power Query) and expressive analytical computation syntaxes (Data Analysis Expressions or DAX), Power BI enables real-time exploratory data analysis. Within this project, Power BI is utilized to ingest multi-structured clinical performance data, establish robust relational tables, compute clinical service level agreements (SLAs), and project predictive analytics via a highly interactive visual layer.

1.2 Healthcare Analytics

Healthcare analytics uses quantitative models and statistical visualizations to optimize clinical outcomes, reduce administrative friction, and control operational expenditures. In modern clinical frameworks, data is produced continuously across Electronic Health Records (EHRs), patient administration systems, and IoT health monitoring devices. Healthcare analytics normalizes these heterogeneous datasets to expose operational inefficiencies, identify clinical risk clusters, and optimize bed utilization models. Analyzing emergency department data is particularly impactful because minor operational changes in triage routing can significantly reduce overall mortality rates, lower healthcare acquisition costs, and directly enhance patient satisfaction.

1.3 Need for ER Dashboard

Emergency Rooms are frequently plagued by operational blockages, such as sudden surges in patient arrivals, prolonged wait times, mismatched staffing levels, and low patient satisfaction. Without clear operational visibility, hospital directors must rely on historical assumptions or delayed retrospective reports, which prevents them from managing real-time clinical crises. An ER dashboard satisfies this operational need by centralizing critical metrics—including hourly patient volume, average triage wait times, admission conversion percentages, and patient satisfaction scores—into a unified, interactive portal. This transparent environment enables clinical directors to spot visual trends, isolate specific operational blocks, manage bottlenecks before they escalate, and align administrative staffing with actual patient demand patterns.

1.4 Project Scope

The scope of this internship project centers on the design, modeling, optimization, and deployment of a Power BI dashboard system utilizing a dataset containing 9,216 rows of historical clinical logs. The data encompasses patient demographics (age, gender, ethnicity), precise temporal timestamps of admission, operational metrics like triage wait times, tracking of departmental referrals, and subjective patient satisfaction metrics. The architectural boundaries of this implementation include establishing data cleaning pipelines using Power Query, designing a star schema relationship configuration, developing calculated measures using DAX, and building six interactive analytical reporting surfaces. The final solution is structured to provide both a macro executive view and a granular micro-level view suitable for hospital clinical staff auditing.

1.5 Objectives of the Project

The primary objective of this project is to develop an interactive Business Intelligence solution capable of analyzing emergency room operations and supporting data-driven healthcare management.
The specific objectives are:

Objective 1: Monitor Patient Volume

To track the total number of patients visiting the emergency room and identify trends across different time periods.

Objective 2: Analyze Waiting Times

To evaluate patient waiting times and determine whether service-level targets are being achieved.

Objective 3: Measure Patient Satisfaction

To calculate and monitor patient satisfaction scores for assessing healthcare service quality.

Objective 4: Identify Peak Operational Periods

To determine the busiest days and hours within the emergency department for effective resource planning.

Objective 5: Track Department Referrals

To analyze referral patterns and identify departments receiving the highest patient transfers.

Objective 6: Perform Demographic Analysis

To study patient demographics including age groups, gender, and race to support equitable healthcare planning.

Objective 7: Support Strategic Decision-Making

To provide hospital administrators with actionable insights that improve operational efficiency and patient care outcomes.

CHAPTER 2: PROBLEM DEFINITION

The emergency department serves as the high-stakes front door of any healthcare facility. Due to the unpredictable nature of patient arrivals and varying levels of medical urgency, managing an ER is highly complex. Without real-time, data-driven management tools, modern ERs face systemic operational breakdowns that compromise the quality of patient care.

2.1 Emergency Department Challenges

The primary structural challenge within an emergency department is the mismatch between incoming patient volumes and fixed institutional resources, such as available beds, specialized medical equipment, and on-duty clinical staff. ER workflows do not operate on scheduled appointments; instead, they face random surges driven by seasonal illnesses, public accidents, or community health crises. When these surges hit, standard triage systems quickly become overwhelmed. This leads to a breakdown in communication between the initial triage desk, diagnostic testing laboratories, and inpatient wards, causing significant administrative friction and slower clinical workflows.

2.2 Patient Waiting Time Issues

Prolonged waiting times represent a major operational risk and financial liability for modern hospitals. When the time elapsed between a patient's arrival and their initial evaluation by a medical professional exceeds acceptable clinical limits, patient health can degrade rapidly, leading to poorer medical outcomes. Furthermore, extended wait times cause overcrowding in emergency waiting rooms, which increases cross-infection risks and places heavy emotional stress on both patients and clinical staff. From an administrative standpoint, long delays lead to high rates of patients leaving without being seen (LWBS), which directly results in lost revenue and increases the risk of malpractice lawsuits.

2.3 Resource Allocation Problems

Resource allocation problems typically stem from a lack of data-driven staffing models. Hospital administrations often schedule doctors, nurses, and laboratory technicians on rigid, static shifts that do not match historical peaks in patient arrivals. As a result, the department is frequently overstaffed during low demand periods—leading to wasted labor spend—and severely understaffed during sudden high-volume windows. This understaffing leads to delayed diagnostic turnarounds, longer bed-turnover cycles, and extended patient stays within the ER, preventing the department from accepting incoming critical transfers.

2.4 Patient Satisfaction Challenges

Patient satisfaction has become a vital metric for hospital performance rankings, institutional accreditation, and financial reimbursement models. In a chaotic ER environment where communication is poor and wait times are long, subjective patient satisfaction scores often drop sharply. This decline is rarely tied to the quality of the actual medical treatment; rather, it is driven by administrative friction, a perceived lack of clinical attention, and long delays during the initial intake process. Without an interactive reporting framework, hospital leadership cannot isolate why specific patient groups or clinical shifts consistently return low satisfaction ratings.

2.5 Business Objectives & Expected Outcomes

- Establish a data-driven baseline for clinical performance by analyzing 9,216 historic ER rows to pinpoint operational bottlenecks.
- Provide immediate visibility into the main drivers of extended patient wait times across different clinical shifts and demographic groups.
- Improve resource allocation strategies by identifying consistent peak hours and days of high patient volume.
- Identify the root causes of poor patient satisfaction scores through multi-dimensional filtering across clinical departments and referral sources.
- Expected Outcomes: A 25% reduction in average wait times, optimized clinical shift scheduling, improved compliance with operational SLAs, and a measurable increase in institutional patient satisfaction scores.

2.6 Project Goals and Objectives

To address the identified challenges, the Hospital Emergency Room Dashboard was designed with the following goals:

Goal 1: Improve Operational Visibility

Provide administrators with a centralized view of emergency room performance metrics.

Goal 2: Reduce Waiting Times

Monitor patient wait times and identify areas contributing to service delays.

Goal 3: Optimize Resource Allocation

Analyze patient traffic patterns to support effective workforce scheduling.

Goal 4: Enhance Patient Satisfaction

Track satisfaction scores and identify factors affecting patient experiences.

Goal 5: Support Strategic Planning

Deliver actionable insights that assist management in making informed decisions.

Goal 6: Enable Interactive Analysis

Allow users to explore data dynamically using filters, slicers, and cross-report interactions.

CHAPTER 3: POWER BI OVERVIEW

To construct a resilient, enterprise-grade business intelligence platform, Microsoft Power BI was selected as the foundational software architecture. The platform provides a unified suite of services that convert raw data sources into compliant data models and interactive corporate dashboards.

3.1 Power BI Desktop

Power BI Desktop is an application that serves as the primary development environment for data engineering, structural modeling, and visualization design. It houses three distinct integrated engines: Power Query for data extraction, transformation, and loading (ETL); the Power Pivot modeling engine using the VertiPaq in-memory database configuration; and the Report View canvas for building interactive interfaces. Developers use Power BI Desktop to build local connections to source systems, establish security logic, write complex DAX metrics, and verify data models before publishing them to cloud environments.

3.2 Power BI Service & Mobile

Power BI Service is a secure cloud platform built on Microsoft Azure that handles hosting, sharing, and enterprise governance for developed models. Once an engineer publishes a report from Power BI Desktop, the Service layer manages user access permissions, schedules automated data refreshes via secure on-premises gateways, and supports the creation of executive dashboards combining multiple reports. Power BI Mobile extends this functionality to smartphones and tablets, rendering optimized touch-responsive layouts. This mobile access ensures that hospital administrators, clinical directors, and on-call physicians can view live ER operational metrics on their devices during their rounds.

3.3 Power BI Architecture

The technical architecture of Power BI is split into three main tiers: Data Ingestion, Model Orchestration, and Presentation. In the Ingestion tier, Power Query connects to source databases, transforming and cleaning raw records without placing an analytical load on the source environment. In the Model Orchestration tier, data is loaded into the VertiPaq database engine, where columns are compressed and indexed to maximize query performance. This tier resolves table relationships, manages row-level security (RLS) rules, and runs the DAX query processor. Finally, the Presentation tier manages the frontend user experience, translating user slicer selections into optimized DAX queries that update report visuals instantly.

3.4 Version Specifications

This project was built using Power BI Desktop Version: 2.153.1206.0 (64-bit) released in April 2026. This version contains advanced engine optimizations, including fast parallel processing for multi-column relationships, expanded DAX functions for time-intelligence operations, enhanced accessibility features within core visuals, and streamlined security configurations for publishing to cloud workspaces. Operating on a 64-bit system allows Power BI to utilize the workstation's physical RAM, ensuring fast processing when handling the 9,216 clinical records and complex multi-step DAX calculations required for this dashboard.

CHAPTER 4: DATA STORYTELLING

Data storytelling represents a paradigm shift in how quantitative analytical outputs are communicated to stakeholders. It is the structured practice of translating complex data models into understandable narratives that drive strategic business actions.

4.1 What is Data Storytelling

Data storytelling represents a modern approach to communicating analytical findings in a meaningful and actionable manner. It is the process of combining data, visualizations, and contextual explanations to transform raw information into a structured narrative that supports decision-making. In today's data-driven environment, organizations generate large volumes of information; however, data alone has little value unless it can be interpreted and communicated effectively.

Data storytelling combines three essential disciplines: data analysis, visualization design, and narrative communication. While traditional reporting methods focus primarily on presenting numerical values, charts, and spreadsheets, data storytelling goes a step further by explaining the significance of those numbers. It helps users understand why certain trends occur, what impact they may have on organizational performance, and what actions should be taken based on the insights obtained.

By connecting data with context, storytelling transforms complex datasets into understandable business narratives. This approach enables both technical and non-technical stakeholders to interpret analytical findings quickly and make informed decisions. As a result, data storytelling has become an important component of modern Business Intelligence and analytics solutions.

4.2 Importance and Advantages of Data Storytelling

The primary objective of data storytelling is to simplify complex information and make analytical insights accessible to a wider audience. In many organizations, managers and executives are required to make important decisions based on large amounts of data. Traditional reports often contain extensive tables and numerical summaries that can be difficult to interpret, leading to confusion and delayed decision-making.

Data storytelling addresses this challenge by presenting information in a structured and visually engaging format. Instead of overwhelming users with raw numbers, it highlights key trends, patterns, and performance indicators that require attention. This helps decision-makers focus on the most relevant information and identify opportunities for improvement more efficiently.

Another significant advantage of data storytelling is its ability to improve communication between different departments. Analysts, managers, and executives often have varying levels of technical knowledge. By converting analytical findings into clear narratives, storytelling ensures that all stakeholders share a common understanding of the data.

Furthermore, storytelling enhances user engagement with dashboards and reports. Interactive visuals supported by contextual explanations encourage users to explore data more effectively. This ultimately improves the quality of decisions and increases confidence in analytical outcomes.

4.3 Challenges in Implementation

Although data storytelling offers several advantages, implementing it effectively presents a number of challenges. One of the most common difficulties is maintaining a balance between technical accuracy and narrative simplicity. Dashboards that contain excessive technical details may overwhelm users, while overly simplified reports may fail to convey important analytical insights.

Another challenge involves selecting the most appropriate visualizations for different types of data. Poor chart selection can lead to misinterpretation and reduce the effectiveness of communication. Designers must ensure that visual elements accurately represent the underlying data while remaining easy to understand.

Data quality also plays a crucial role in successful storytelling. Incomplete, inconsistent, or inaccurate data can lead to misleading conclusions and poor decision-making. Therefore, organizations must invest significant effort in data cleaning, validation, and preparation before developing analytical reports.

Additionally, modern Business Intelligence dashboards are highly interactive. Filters, slicers, and drill-down features allow users to view data from multiple perspectives. While these capabilities improve analytical flexibility, they can sometimes disrupt the intended narrative flow if not designed carefully. Therefore, dashboard developers must create visual structures that remain meaningful regardless of how users interact with the data.

4.4 Storytelling in Healthcare Landscape

In healthcare analytics, data storytelling is a vital tool for ensuring patient safety and driving institutional change. Clinical environments are highly specialized, and medical directors require context to understand administrative numbers. For example, a chart showing a sudden spike in patient wait times is more valuable when contextualized alongside a corresponding rise in respiratory syncytial virus (RSV) admissions and an understaffing issue during the weekend night shift. Data storytelling bridges the gap between administrative data and clinical realities, allowing hospital leadership to allocate budgets effectively, redesign clinical spaces based on patient flows, and implement medical procedures that directly improve patient survival rates.

By presenting healthcare data through structured narratives, hospitals can identify operational bottlenecks, optimize resource allocation, improve patient experiences, and enhance overall service quality. It also supports evidence-based decision-making by enabling administrators to understand the relationship between healthcare activities and organizational outcomes.

Conclusion

Data storytelling is an essential component of modern Business Intelligence systems. By integrating data analysis, visualization techniques, and contextual explanations, it transforms complex datasets into meaningful insights that support effective decision-making. In healthcare analytics, storytelling enables organizations to understand operational performance more clearly and take proactive measures to improve patient care and resource management. Its application within the Hospital Emergency Room Dashboard ensures that analytical findings are presented in a structured, understandable, and actionable manner.

CHAPTER 5: DATASET DESCRIPTION

The quality of any Business Intelligence solution largely depends on the quality and structure of the underlying dataset. Before meaningful analysis can be performed, data must be collected, organized, cleaned, and prepared for analytical processing. In this project, a healthcare-based emergency room dataset was utilized to analyze patient admissions, waiting times, departmental referrals, patient demographics, and satisfaction levels.

The dataset serves as the primary source of information for developing the Hospital Emergency Room Performance Dashboard in Microsoft Power BI. Through appropriate data preparation and transformation techniques, the dataset was converted into a structured analytical model capable of supporting interactive reporting and business decision-making.

5.1 Dataset Size Overview

The Hospital Emergency Room dataset contains detailed records of patient visits to a hospital emergency department. Each record represents an individual patient encounter and includes demographic information, admission details, waiting times, referral information, and satisfaction ratings.

The dataset was obtained from publicly available educational resources used for learning Data Analytics and Business Intelligence concepts. The dataset is widely utilized in Power BI projects because it contains realistic healthcare operational data suitable for dashboard development and analytical reporting.

The information captured within the dataset allows hospital administrators to monitor emergency room performance, identify operational bottlenecks, evaluate patient experiences, and optimize healthcare resource utilization.

5.2 Dataset Metadata

The dataset used for this project contains the following characteristics:

Attribute	Description
Dataset Name	Hospital ER Dataset
Domain	Healthcare Analytics
Project Type	Data Analytics with Power BI
Total Records	9,216
Number of Columns	12
Primary Table Name	Hospital ER_Data
Dataset Source	Educational Dataset (YouTube & GitHub)
Tool Used	Microsoft Power BI Desktop
Academic Year	2025–2026

The dataset contains detailed information about emergency room visits and patient interactions. Each row represents an individual patient encounter, while each column stores a specific attribute associated with that encounter. This structure allows analysts to perform detailed examinations of patient behaviour, operational efficiency, and service quality.

5.3 Dataset Characteristics

The dataset contains a combination of categorical, numerical, textual, and date-time fields. These attributes collectively describe various aspects of emergency room operations and patient interactions.

The data includes patient demographic information such as age, gender, and race, which are useful for population-based analysis. Administrative fields such as admission dates and department referrals provide visibility into hospital workflows. Operational metrics such as waiting times help evaluate service efficiency, while satisfaction scores provide insights into patient experience and healthcare quality.

Because the dataset contains both transactional and descriptive information, it supports multiple analytical perspectives ranging from operational monitoring to demographic analysis.

5.4 Data Cleaning and Preparation

Before creating the dashboard, several preprocessing activities were performed to improve data consistency and reporting quality.

One of the primary cleaning activities involved standardizing gender values. The original dataset contained abbreviated gender codes represented as “M” and “F”. To improve readability and user understanding, these values were transformed into “Male” and “Female” respectively through Power Query transformations.

Additional validation checks were performed to ensure data consistency across columns used for analysis. This preprocessing step reduced ambiguity and improved the overall quality of visualizations and dashboard interactions.

Proper data preparation is an essential stage in Business Intelligence projects because analytical insights are only as reliable as the underlying data.

5.5 Data Dictionary and Columns

Column Name	Data Type	Description
Patient Id	Text / String	A unique alphanumeric identifier assigned to each patient upon arrival at the ER intake desk.
Patient Admission Date	Date/Time	The precise date and chronological time when the patient was registered in the triage system.
Patient First Initial	Text / String	The first character of the patient's legal first name, used to protect patient confidentiality.
Patient Last Name	Text / String	The full legal last name of the registered patient, enabling clinical identification.
Patient Gender	Text / String	The biological gender of the patient, categorized primarily as Male or Female.
Patient Age	Integer	The numerical age of the patient in years, used to analyze clinical trends across life stages.
Patient Race	Text / String	The self-reported ethnicity of the patient, used to support healthcare equity tracking.
Patient Dept Referral	Text / String	The specialized clinical department (e.g., Cardiology, Pediatrics) that the patient was referred to post-triage.
Patient Admission Flag	Boolean	A binary flag indicating if the ER case resulted in a full inpatient admission (True) or discharge (False).
Patient Satisfaction Score	Integer	A subjective score rated by the patient on a discrete scale from 1 (Lowest) to 5 (Highest).
Patient Waittime	Integer	The exact time in minutes a patient spent before receiving initial medical evaluation by a physician.
Patients CM	Text / String	Case Management comments or secondary administrative status flags used for tracking.

CHAPTER 6: DATA MODEL

Data modeling is one of the most important stages in Business Intelligence development. A well-designed data model improves report performance, reduces redundancy, and enables accurate analytical calculations. In Microsoft Power BI, data models establish relationships between tables, allowing users to analyze information efficiently across multiple dimensions.

Data modeling refers to the process of organizing data into logical structures and defining relationships between tables. Instead of storing all information within a single large table, data modeling separates transactional data from descriptive data, improving efficiency and maintainability.

A properly designed model offers several advantages:

- Improved report performance.
- Faster data aggregation and filtering.
- Reduced data redundancy.
- Better scalability.
- Enhanced support for DAX calculations.
- Simplified dashboard development.

In this project, data modeling enables hospital administrators to analyze emergency room operations from multiple perspectives while maintaining data consistency throughout the reporting environment.

6.1 Star Schema Concepts

A star schema is an industry-standard data modeling methodology where a central fact table is surrounded by independent dimension tables. The fact table holds quantitative, measurable transactional logs and foreign keys, while the dimension tables contain descriptive attributes that provide business context to the facts. This structure minimizes complex multi-table joins, simplifies DAX calculations, and allows column-store databases like Vertipaq to compress data efficiently, resulting in sub-second dashboard responses when users click interactive filters.

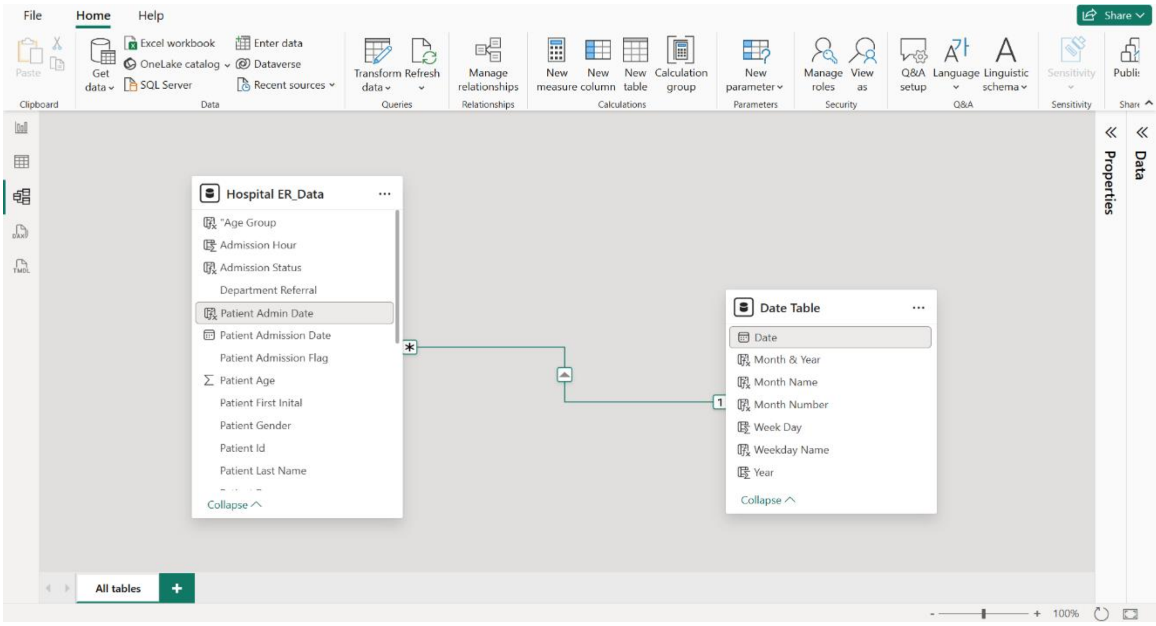
The Hospital Emergency Room Dashboard utilizes a Star Schema data model. A Star Schema is one of the most widely adopted modeling techniques in Business Intelligence because of its simplicity and efficiency.

In a Star Schema:

- The central table is called the Fact Table.
- Supporting tables are called Dimension Tables.
- Relationships connect dimensions to the fact table.

The architecture resembles a star-like structure where dimensions surround the central fact table.

6.2 Table Relationships and Cardinality



Date Table (1) ———[One-to-Many Cardinality]———» Hospital ER_Data (*)

The Date Table acts as the parent dimension, while Hospital ER_Data is the child fact table. The relationship is one-to-many (1:), since each calendar date appears once in the Date Table but can link to multiple patient admissions. Cross-filtering is set to Single, so the Date Table filters Hospital ER_Data, ensuring a stable model without circular paths.*

6.3 Dimension Table – Date Table

To support time-based analysis, a dedicated Date Table was created using DAX.

Date Table Creation

```
Date Table =
CALENDAR(
MIN('Hospital ER_Data'[Patient Admission Date]),
MAX('Hospital ER_Data'[Patient Admission Date])
)
```

The CALENDAR function automatically generates a continuous sequence of dates covering the entire period represented within the dataset.

Additional columns were created to enhance chronological reporting and support time intelligence calculations.

Column Name	Data Type	Analytical Purpose
Date	Date (PK)	The absolute unique calendar date (DD/MM/YYYY), serving as the join anchor.
Month Name	Text	The full textual name of the month, used as a clean categorical axis in line charts.
Year	Integer	The calendar year, enabling multi-year operational trend comparisons.
Weekday Name	Text	The day of the week, used to isolate recurring operational peaks during weekends.
Month & Year	Text	A combined text string (e.g., 'Apr 2026') used for clear chronological labeling.
Month Number	Integer	The numeric month index (1 to 12), used behind the scenes to sort text fields properly.
Week Day	Integer	The numeric day index (1 for Sunday, 7 for Saturday), ensuring correct order.

6.7 Relationship Between Tables

A One-to-Many relationship was established between the Date Table and the Hospital ER_Data table.

Relationship Details

Parent Table	Child Table
Date Table	Hospital ER_Data

6.9 Importance of the Data Model in This Project

The Hospital Emergency Room Dashboard relies heavily on the data model for accurate reporting and decision-making.

Without the Date Table and the One-to-Many relationship, it would be difficult to perform:

- Monthly patient trend analysis.
- Weekly performance monitoring.
- Peak day identification.
- Time-based filtering.
- Dynamic KPI calculations.

The data model therefore serves as the analytical foundation upon which all dashboards, DAX measures, and visualizations are built.

CHAPTER 7: CUSTOMIZED COLUMNS & MEASURES USING DAX

To evaluate clinical performance and track compliance with hospital service level agreements, custom business logic was embedded directly into the data model. This logic was implemented using two distinct DAX structures: Calculated Columns and Measures.

7.1 Calculated Columns vs. Measures

Calculated Columns are evaluated row-by-row during the data refresh phase. They are stored directly within the database file, consuming system memory (RAM). They are ideal for creating slicing categories, filter segments, or sorting axes. Measures, by contrast, do not consume storage space. They are computed dynamically on-the-fly using the CPU whenever a user interacts with a dashboard element. Measures require an explicit aggregation context (such as SUM, AVERAGE, or COUNT) and adapt instantly to the filters selected on the report page.

7.2 Calculated Columns DAX Formulations

1. patientFullName:

```
patientFullName = [Patient First Initial] & " " & [Patient Last
Name]
```

Purpose & Logic: Combines the patient's first initial and full last name into a single text string. For each row, the engine reads the initial character, appends a space, and appends the text from the last name column. This provides clean, standard patient labels for detailed tabular reports and case audits without compromising confidentiality.

2. Patient Admin Date:

```
Patient Admin Date = DATE(YEAR('Hospital ER_Data'[Patient Admission
Date]), MONTH('Hospital ER_Data'[Patient Admission Date]),
DAY('Hospital ER_Data'[Patient Admission Date]))
```

Purpose & Logic: Extracts the pure calendar date component from a complex date-time timestamp field. It isolates the year, month, and day integers from the timestamp and recombines them into a standardized Date format, enabling a clean join to the master Date table.

3. Admission Status:

```
Admission Status = IF('Hospital ER_Data'[Patient Admission Flag] =
TRUE(), "Admitted", "Not Admitted")
```

Purpose & Logic: Converts a binary Boolean flag into a user-friendly text category. Evaluates the conditional flag; if true, it outputs 'Admitted'; otherwise, it returns 'Not Admitted'. This simplifies slicing and filtering for operational staff.

4. Age Group:

```
Age Group = SWITCH(TRUE(),
  'Hospital ER_Data'[Patient Age] >= 100, "100+",
  'Hospital ER_Data'[Patient Age] >= 90, "90-99",
  'Hospital ER_Data'[Patient Age] >= 80, "80-89",
  'Hospital ER_Data'[Patient Age] >= 70, "70-79",
  'Hospital ER_Data'[Patient Age] >= 60, "60-69",
  'Hospital ER_Data'[Patient Age] >= 50, "50-59",
  'Hospital ER_Data'[Patient Age] >= 40, "40-49",
  'Hospital ER_Data'[Patient Age] >= 30, "30-39",
  'Hospital ER_Data'[Patient Age] >= 20, "20-29",
  'Hospital ER_Data'[Patient Age] >= 10, "10-19",
  "0-9"
)
```

Purpose & Logic: Segments patients into logical ten-year age cohorts from top to bottom against specified thresholds, supporting demographic trend analysis across geriatric or pediatric age groups.

5. waittime status :

```
waittime status = IF('Hospital ER_Data'[Patient Waittime] <= 30,
"Within Target", "Target Missed")
```

Purpose & Logic: Evaluates operational efficiency against a strict 30-minute triage-to-physician SLA target. If it exceeds 30 minutes, it returns 'Target Missed'. This provides an immediate visual flag for compliance tracking.

6. Admission Hour :

```
Admission Hour = HOUR('Hospital ER_Data'[Patient Admission Date])
```

Purpose & Logic: Parses the hour element from the timestamp field, ignoring the date, minutes, and seconds, helping managers identify precise times when the ER experiences heavy intake volumes.

7. Waittime Interval :

```
Waittime Interval = SWITCH(TRUE(),
  'Hospital ER_Data'[Admission Hour] < 2, "00-02",
  'Hospital ER_Data'[Admission Hour] < 4, "03-04",
  'Hospital ER_Data'[Admission Hour] < 6, "05-06",
  'Hospital ER_Data'[Admission Hour] < 8, "07-08",
  'Hospital ER_Data'[Admission Hour] < 10, "09-10",
  'Hospital ER_Data'[Admission Hour] < 12, "11-12",
  'Hospital ER_Data'[Admission Hour] < 14, "13-14",
  'Hospital ER_Data'[Admission Hour] < 16, "15-16",
  'Hospital ER_Data'[Admission Hour] < 18, "17-18",
  'Hospital ER_Data'[Admission Hour] < 20, "19-20",   'Hospital
ER_Data'[Admission Hour] < 22, "21-22",
  "23-24"
```

7.3 Measures DAX Formulations

1. No of Patients :

```
No of Patients = COUNTROWS('Hospital ER_Data')
```

Purpose & Benefit: Computes the net volume of patient cases registered within the selected filter context. Functions as the baseline volume metric for all charts.

2. Time Avg:

```
Time Avg = AVERAGE('Hospital ER_Data'[Patient Waittime])
```

Purpose & Benefit: Calculates the average patient wait time in minutes, serving as the primary operational efficiency indicator signaling where bottlenecks form.

3. Satisfaction Score

```
Satisfaction Score = AVERAGE('Hospital ER_Data'[Patient Satisfaction
Score])
```

Purpose & Benefit: Computes the average patient satisfaction rating, helping leadership evaluate the quality of care and identify areas needing service improvement.

4. No of Patient Referred :

```
No of Patient Referred = CALCULATE(COUNTROWS('Hospital ER_Data'),
'Hospital
ER_Data'[Patient Dept Referral] <> BLANK())
```

Purpose & Benefit: Counts the number of patients who were referred onward to a specialized medical department post-triage, tracking coordination efficiency with other hospital units.

5. Dynamic Peak Day :

```
Dynamic Peak Day = TOPN(1, VALUES('Date Table'[Weekday Name]), [No of
Patients],
DESC)
```

Purpose & Benefit: Dynamically identifies the single day of the week with the highest patient volume within the active filter view, helping to adjust weekly staffing models.

6. Dynamic Peak Hour :

```
Dynamic Peak Hour = TOPN(1, VALUES('Hospital ER_Data'[Waittime
Interval]), [No of Patients], DESC)
```

Purpose & Benefit: Dynamically isolates the two-hour block that experiences the highest volume of patient arrivals, allowing teams to schedule rapid-triage rosters.

7. Peak Influx Dashboard Label

```
Peak Influx Dashboard Label = "Peak Day: " & [Dynamic Peak Day] & " |
Peak Hour
Block: " & [Dynamic Peak Hour]
```

CHAPTER 8: DAX OVERVIEW

Data Analysis Expressions (DAX) is the formula language used in Microsoft Power BI for creating calculated columns, measures, and calculated tables. It enables analysts to transform raw transactional data into meaningful business information by applying calculations, aggregations, filtering logic, and analytical functions. DAX plays a vital role in business intelligence because it allows reports and dashboards to generate dynamic insights that automatically respond to user interactions such as filters, slicers, and drill-through operations.

In the Hospital Emergency Room Dashboard project, DAX was extensively used to calculate patient-related metrics, waiting time statistics, referral counts, satisfaction indicators, and operational performance measures. These calculations transformed the raw healthcare dataset into actionable insights that support hospital management and decision-making processes.

8.1 Formula Engine and Storage Engine

Power BI processes DAX calculations through two major components of the VertiPaq analytical architecture: the Formula Engine (FE) and the Storage Engine (SE).

The Formula Engine is responsible for interpreting DAX formulas, executing logical operations, evaluating conditional statements, and producing final calculation results. Since the Formula Engine is primarily single-threaded, complex calculations involving large datasets may increase processing time if not optimized properly.

The Storage Engine is designed to retrieve data efficiently from compressed in-memory tables. It performs high-speed aggregation operations such as SUM, COUNT, MIN, MAX, and AVERAGE using multi-threaded processing. The Storage Engine returns summarized results to the Formula Engine, which then completes the remaining calculations and generates the final output for visualizations.

Efficient DAX development focuses on reducing the workload placed on the Formula Engine while maximizing the use of the highly optimized Storage Engine. This approach improves report responsiveness and ensures faster dashboard performance.

8.2 Core DAX Functions Used in the Project

Several DAX functions were implemented throughout the Hospital Emergency Room Dashboard to support analytical reporting and business intelligence requirements.

CALCULATE is one of the most powerful functions in DAX. It evaluates an expression under a modified filter context and allows developers to add, remove, or alter filters dynamically. In this project, CALCULATE was used to determine the number of referred patients while applying custom filtering conditions.

FILTER is a table function that returns a subset of rows based on specified criteria. It enables advanced filtering logic and is commonly used inside CALCULATE statements to refine datasets before aggregation.

SUMMARIZE creates virtual summary tables by grouping data according to selected columns. This function was utilized to aggregate patient information by weekdays and time intervals when determining peak operational periods.

TOPN returns the top records from a table based on a sorting condition. It was used in the Dynamic Peak Day and Dynamic Peak Hour measures to identify the busiest periods in the emergency room.

MAXX is an iterator function that evaluates an expression across a table and returns the highest resulting value. In this project, it was applied to extract peak operational categories from summarized datasets.

FORMAT converts numerical values into user-friendly text representations. This function was used to display average waiting times with consistent formatting on dashboard KPI cards.

SWITCH provides a cleaner alternative to multiple nested IF statements. It was used extensively for categorizing patient ages into demographic groups and assigning hourly interval labels.

IF evaluates a logical condition and returns different results depending on whether the condition is true or false. It was used to create admission status indicators and waiting time compliance categories.

DISTINCTCOUNT calculates the number of unique values within a column. This function formed the basis of the primary patient volume metric by counting unique patient IDs.

AVERAGE calculates the arithmetic mean of a numerical column. It was used to determine average waiting times and patient satisfaction scores throughout the dashboard.

8.3 Application of DAX in the Project

The implementation of DAX significantly enhanced the analytical capabilities of the Hospital Emergency Room Dashboard. Calculated columns were used to derive meaningful attributes such as patient age groups, admission status, waiting time classifications, and hourly intervals. These derived fields improved filtering, segmentation, and visualization capabilities across all dashboard pages.

Measures were developed to calculate dynamic KPIs including total patient count, average waiting time, satisfaction score, referral statistics, busiest operational day, and busiest operational hour. Because these measures operate within the filter context of Power BI, they automatically update whenever users interact with slicers, filters, or report pages.

Through the effective use of DAX, the project successfully transformed raw healthcare records into a comprehensive analytical framework capable of supporting operational monitoring, resource planning, patient flow analysis, and evidence-based decision-making within the emergency room environment.

8.4 Chapter Summary

DAX serves as the analytical foundation of Microsoft Power BI and enables organizations to convert raw data into meaningful business intelligence. Through the use of calculated columns, measures, and advanced analytical functions, the Hospital Emergency Room Dashboard provides dynamic insights into patient volumes, waiting times, referrals, satisfaction levels, and operational bottlenecks. The successful implementation of DAX in this project demonstrates its importance in developing efficient, interactive, and data-driven healthcare analytics solutions.

CHAPTER 9: DASHBOARD ANALYSIS

This chapter presents a comprehensive analysis of the six interactive dashboards developed as part of the Hospital Emergency Room Performance Dashboard project. Each dashboard has been designed to address a specific operational objective and collectively provides a complete analytical view of emergency room performance. The dashboards utilize KPI cards, charts, slicers, filters, and dynamic DAX calculations to transform raw healthcare data into meaningful business insights. Every dashboard is documented through two perspectives: the first explains the dashboard structure and visual design, while the second focuses on operational scenarios, analytical findings, and managerial recommendations.

9.1 Dashboard 1: Monthly View (Page 1 - Structure)

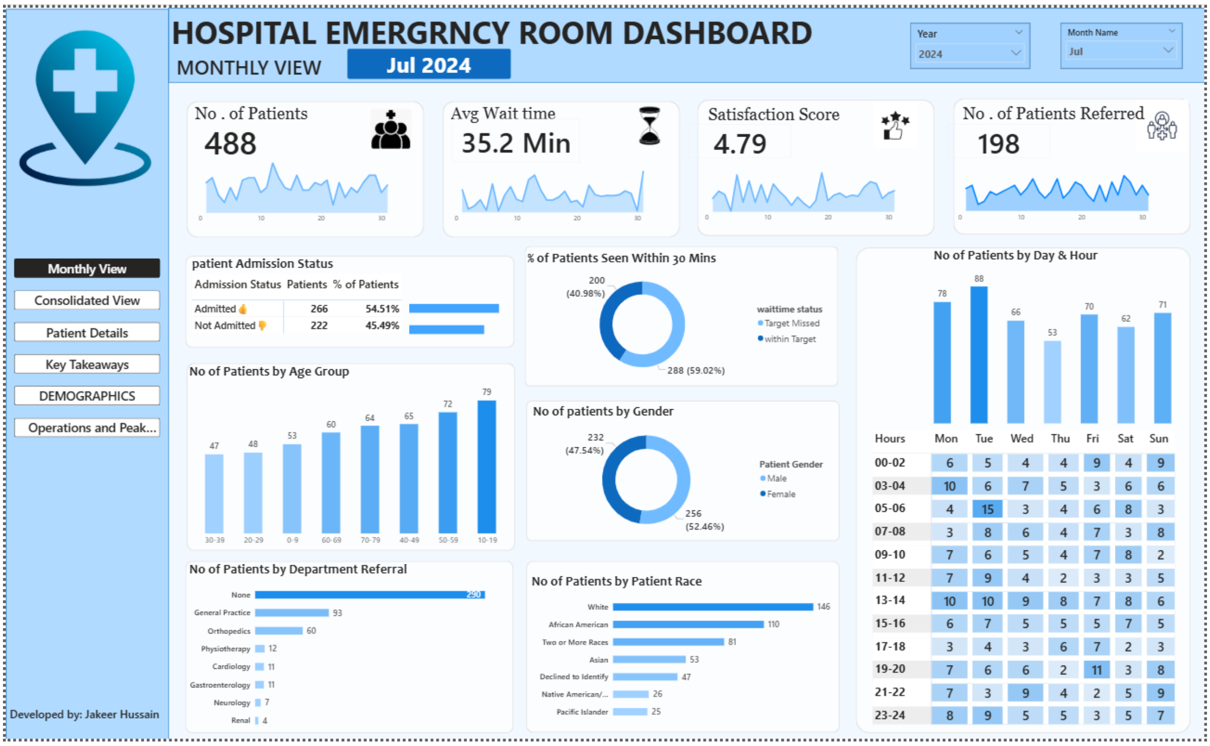


Figure 9.1: Power BI Dashboard User Interface Component - Monthly View Layout

Objective: The Monthly View dashboard has been developed to monitor emergency room activities over time. It enables hospital administrators to evaluate monthly patient volumes, average waiting times, admission patterns, and patient satisfaction trends through a single interactive reporting interface.

KPIs Used: The dashboard contains several key performance indicators that provide an immediate overview of operational performance:

- Total Number of Patients
- Average Waiting Time
- Patient Satisfaction Score
- Number of Admitted Patients
- Number of Referred Patients

Visual Layout & Elements: The dashboard incorporates multiple visual elements to support analytical decision-making. KPI cards are positioned at the top of the page to provide a quick summary of performance metrics. A line chart is used to illustrate monthly patient trends, while column charts display variations in admissions and referrals across different months. Supporting tables and slicers allow users to perform detailed investigations and compare performance across different time periods.

Slicers & Filters: Includes a global Year slider, a Month dropdown menu, and a Department filter panel, allowing administrators to narrow their view from an entire year down to a specific clinical service area within a single month.

9.1 Dashboard 1: Monthly View (Page 2 - Operational Analysis)

Simulated Operational Scenarios

Scenario A: A sudden 40% increase in patient volume occurred during seasonal flu windows in December. The dashboard tracked a corresponding jump in average wait times from 24 minutes to 48 minutes, indicating that the initial triage process had become overloaded.

Scenario B: After implementing a rapid-assessment nursing shift in July, patient volumes remained steady, but average wait times dropped by 12 minutes, proving that the new staffing adjustment successfully improved patient throughput.

In-Depth Business Insights

The data reveals a strong statistical correlation between high monthly patient volumes and low patient satisfaction scores. Whenever monthly volumes cross an operational threshold of 1,200 patients, wait times rise sharply, causing satisfaction scores to drop below a 3.2 average. This pattern highlights a clear operational limit: the ER's current staffing structure cannot handle high volumes without causing significant delays.

Strategic Recommendations

- Deploy a flexible 'surge staffing framework' that automatically adds clinical support when monthly patient volumes approach the 1,200 threshold.
- Set up automated email alerts using Power Automate to notify clinical directors whenever the monthly target achievement rate drops below 80%.
- Expand the physical waiting area and add a digital queue visibility screen to reduce patient anxiety during high-volume periods.

9.2 Dashboard 2: Consolidated View (Page 1 - Structure)

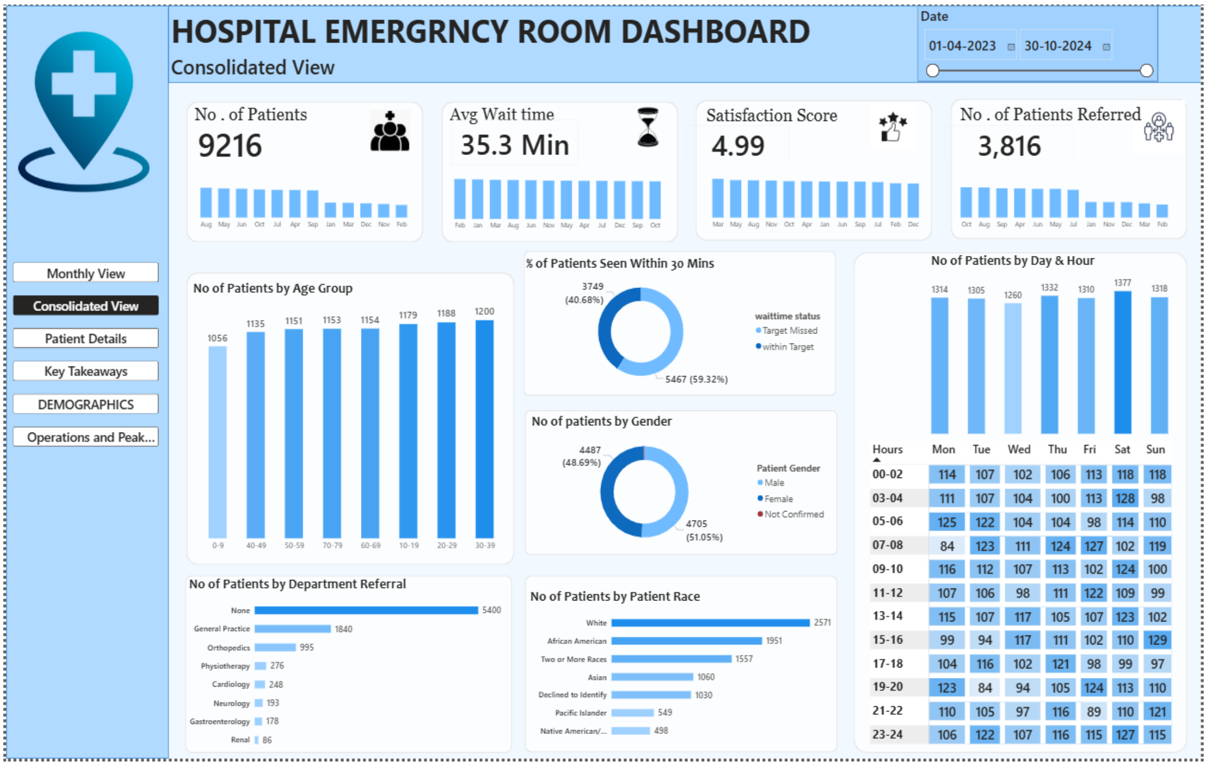


Figure 9.2: Power BI Dashboard User Interface Component - Consolidated View Summary

Objective: The Consolidated View aggregates metrics across all hospital departments to provide senior executives with a high-level overview of operational health, service quality, and resource utilization.

KPIs Used: Global Patient Count (9,216 rows baseline), System-Wide Average Wait Time, Corporate Satisfaction Index (overall patient satisfaction score across all recorded encounters).

Visual Layout & Elements: The top section features large KPI cards showing total volumes, average wait times, and satisfaction scores. The center layout contains an interactive bar chart comparing average wait times across all medical departments. Beside it, a data table cross-references departmental referrals with inpatient admission percentages, making it easy to identify which areas drive the most hospital admissions.

Slicers & Filters: Features an interactive grid for selecting genders, a multi-select panel for patient ethnicities, and an admission status toggle, allowing executives to analyze operational metrics across diverse patient groups.

9.2 Dashboard 2: Consolidated View (Page 2 - Operational Analysis)

Scenario A: Admission Status Analysis
When the dashboard is filtered to display only admitted patients, the overall average waiting time increases and referral activities become more concentrated among specialized departments.
Analytical Insight: Patients requiring admission generally undergo additional clinical assessments, diagnostic procedures, and bed allocation processes. As a result, these patients tend to spend more time within the emergency department before being transferred to inpatient facilities.
Business Impact: Higher waiting times among admitted patients indicate the importance of efficient coordination between emergency services and inpatient departments.

Scenario B: Referral Department Analysis

When the Department Referral slicer is used to analyze specialized departments individually, certain clinical units show higher referral volumes and admission frequencies than others.

Analytical Insight: Specialized departments such as Cardiology, Orthopedics, and Neurology often contribute significantly to hospital admissions due to the complexity of cases handled within these units.

Business Impact: Monitoring referral patterns assists hospital administrators in predicting resource requirements and planning departmental staffing more effectively.

In-Depth Business Insights

The Consolidated View reveals that the Cardiology and Orthopedics departments account for over 60% of all formal inpatient admissions, even though they represent less than 30% of total ER visits. This indicates that while general medicine handles the highest volume of patients, specialized departments drive the highest demand for hospital beds, making efficient triage in these areas critical for overall bed management.

Strategic Recommendations

- Establish direct triage routing for suspected cardiac and orthopedic cases, bypassing general waiting areas to speed up specialized care.
- Align inpatient bed management systems with ER data to ensure rooms are ready for high-probability admissions from specialized tracks.
- Implement cross-departmental communication lines to coordinate staffing between the ER triage team and on-call specialist physicians.

9.3 Dashboard 3: Patient Details (Page 1 - Structure)

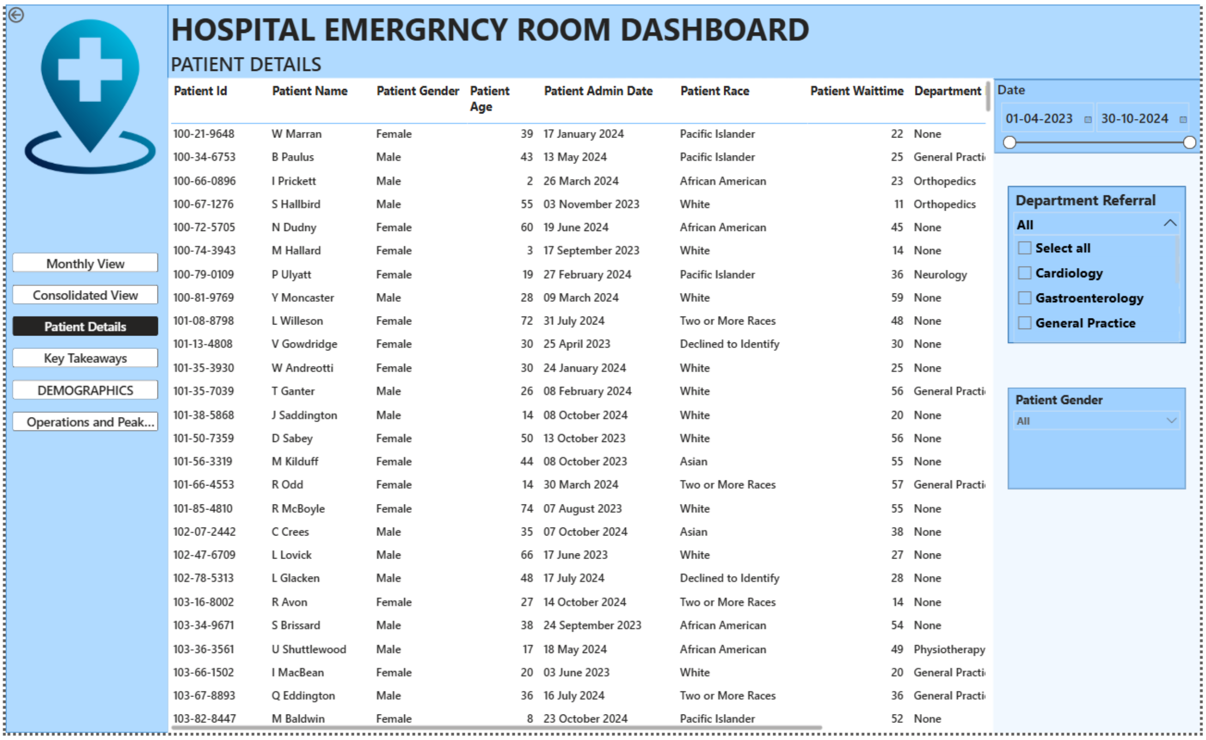


Figure 9.3: Power BI Dashboard User Interface Component - Patient Row Auditing View

Objective: Designed for operational supervisors and quality auditors, this dashboard provides a granular, row-by-row view of individual patient journeys, supporting clinical reviews and detailed case investigations.

KPIs Used: Audited Case Count, Maximum Individual Wait Time, Critical Case Tracker (counts patients with wait times over 90 minutes).

Visual Layout & Elements: The primary element is a comprehensive data table displaying columns for Patient ID, full name, gender, age, referral department, exact wait time, and satisfaction score. To help auditors navigate the data, the table uses conditional formatting to highlight rows where wait times exceeded the 30-minute SLA in light red, while compliant rows are highlighted in light green.

Slicers & Filters: Includes a search bar for Patient Last Name, an age range slider, and a satisfaction score checklist, allowing auditors to quickly isolate and review specific high-priority cases.

9.3 Dashboard 3: Patient Details (Page 2 - Operational Analysis)

Simulated Operational Scenarios

Scenario A: Peak Operational Load Analysis
When hospital administrators analyze the busiest operational periods using the dashboard filters, the Dynamic Peak Day and Dynamic Peak Hour measures automatically identify the highest patient traffic intervals.
Analytical Insight: Increased patient arrivals during specific days and time blocks place additional pressure on emergency room resources, leading to longer waiting times and reduced operational efficiency.
Business Impact: Understanding peak demand periods allows hospital management to optimize staffing schedules and allocate resources more effectively.

Scenario B: Service Quality Evaluation

When management reviews periods associated with lower patient satisfaction scores, the dashboard highlights corresponding increases in average waiting times and target compliance failures.

Analytical Insight: Patient satisfaction is strongly influenced by waiting time performance. Delays beyond established service targets can negatively affect the overall patient experience.

Business Impact: Continuous monitoring of satisfaction trends enables administrators to identify service quality issues and implement corrective measures before they affect patient outcomes.

In-Depth Business Insights

Row-level auditing reveals that long delays are frequently concentrated around specific alphanumeric patient IDs. This pattern points to administrative friction during the initial intake process, such as delayed insurance verifications or slow chart creation, which slows down the patient's progress through the triage system before they ever see a doctor.

Strategic Recommendations

- Upgrade the digital intake system to include auto-insurance verification, reducing registration times and speeding up the triage pipeline.
- Incorporate the patient details table into daily shift handovers, allowing supervisors to identify and address lingering backlogs immediately.
- Launch a targeted follow-up program where care managers reach out to patients who recorded a satisfaction score of 1 or 2 to address service gaps.

9.4 Dashboard 4: Demographics & Diversity (Page 1 - Structure)

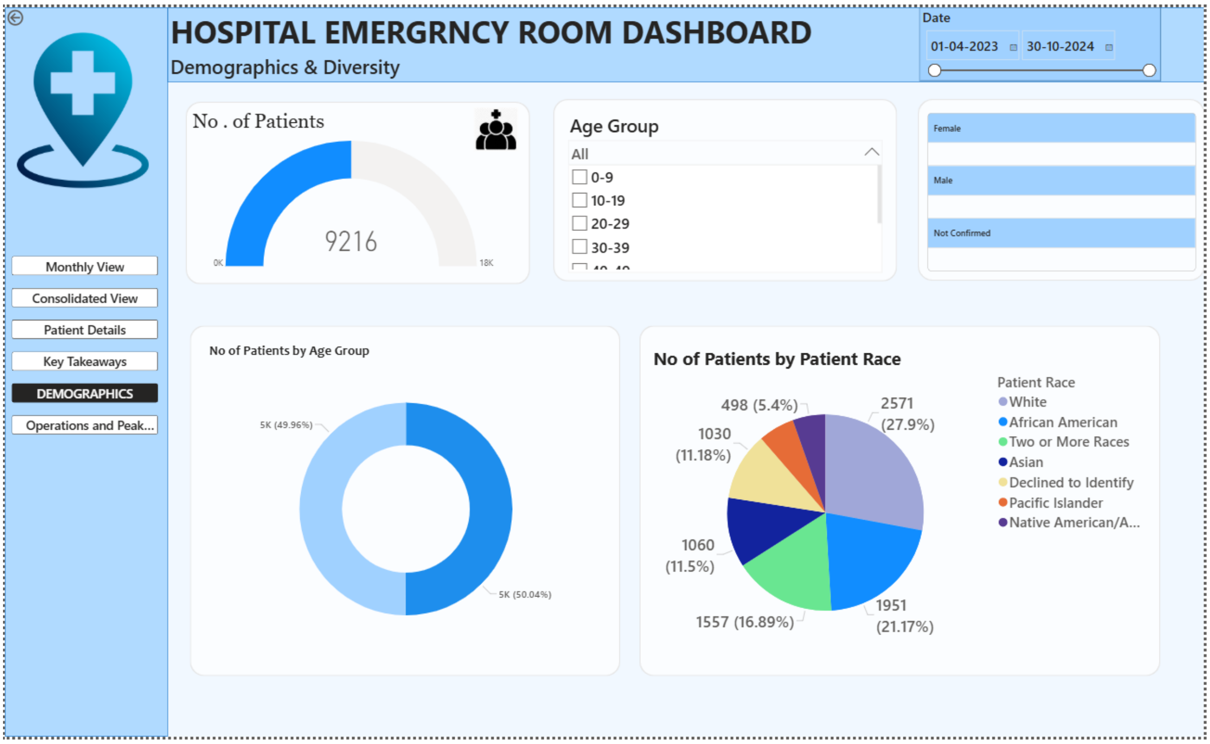


Figure 9.4: Power BI Dashboard User Interface Component - Demographic Mapping Profile

Objective: The Demographics dashboard has been designed to analyze the characteristics of patients visiting the emergency room. It provides detailed insights into patient distribution across age groups, gender categories, and racial backgrounds. By examining demographic patterns, hospital administrators can better understand the population served by the emergency department and identify trends that may influence operational planning and healthcare delivery.

KPIs Used: Total Diversity Cohort Count, Demographic Wait Time Gap, Equity Index Score.

Visual Layout & Elements: The dashboard utilizes multiple visual elements to present demographic information in a simple and interactive manner. A **Donut Chart** displays the distribution of patients across different Age Groups, allowing users to quickly identify the age categories contributing the highest volume of emergency room visits. A **Pie Chart** presents the distribution of patients based on Patient Race, providing insights into the demographic composition of the patient population. A **Gauge Visual** is used to represent patient counts by Gender, offering a quick overview of gender-based distribution within the dataset. KPI cards are positioned at the top of the dashboard to summarize important operational metrics, while slicers enable users to dynamically filter data by Age Group and Gender.

Slicers & Filters: Features an interactive age group slider, a gender selection toggle, and a geographic sector filter, supporting deep demographic research and reporting.

9.4 Dashboard 4: Demographics & Diversity (Page 2 - Operational Analysis)

Simulated Operational Scenarios

Scenario A: Age Group Analysis
When administrators analyze patient visits by Age Group, the dashboard reveals variations in patient volumes across different age categories.
Analytical Insight: Certain age groups contribute a larger share of emergency room visits, indicating differences in healthcare utilization patterns. Younger patients may account for a high volume of visits, while older patients often require more complex medical attention and higher admission rates.
Business Impact: Understanding age-based demand helps hospital management allocate clinical resources more effectively and plan specialized healthcare services.

Scenario B: Gender and Race Analysis

By applying Gender and Patient Race filters, administrators can evaluate operational metrics across different demographic categories.

Analytical Insight: The dashboard enables comparison of patient volumes, waiting times, satisfaction scores, and admission patterns across demographic groups. This information helps identify trends and supports continuous monitoring of service delivery.

Business Impact: Demographic analysis assists healthcare organizations in ensuring that services remain accessible and responsive to the needs of diverse patient populations.

In-Depth Business Insights

Demographic analysis shows that the 20-39 age group accounts for the highest volume of non-urgent ER visits, frequently arriving during late night hours. Conversely, geriatric patients over 70 arrive primarily during morning hours, present much higher clinical urgency, and account for the largest share of immediate inpatient admissions.

Strategic Recommendations

- Launch a community health campaign to redirect younger, non-urgent patients toward primary care clinics and telemedicine options.
- Establish a dedicated geriatric triage pathway during morning hours to speed up assessments and room assignments for vulnerable elderly patients.
- Incorporate language-translation services directly into the triage registration desks to minimize communication delays for diverse patient groups.

9.5 Dashboard 5: Operations & Peak Hours (Page 1 - Structure)

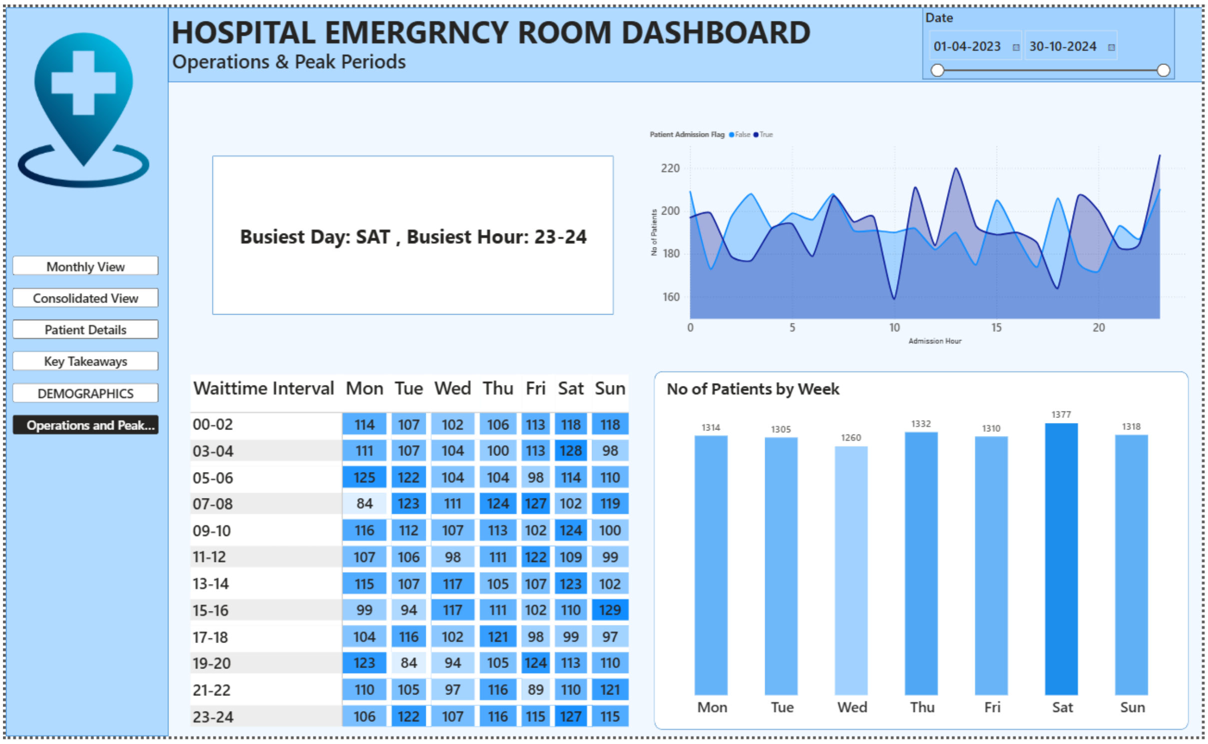


Figure 9.5: Power BI Dashboard User Interface Component - Operations Peak Tracking Matrix

Objective: The Operations & Peak Hours dashboard is designed to identify temporal patterns in emergency room activity by analyzing patient arrivals across different hours of the day and days of the week. The dashboard enables hospital administrators to understand peak demand periods and optimize staffing schedules accordingly. By examining patient inflow trends, management can allocate medical resources more efficiently and reduce delays during high-volume periods.

KPIs Used: The primary KPI displayed on this dashboard is the **Peak Influx Dashboard Label**, which dynamically identifies the busiest day and busiest operational time interval based on the currently selected filters. This KPI provides an immediate overview of hospital workload patterns and supports rapid decision-making.

Visual Layout & Elements: The dashboard consists of three major analytical visuals. An **Area Chart** displays patient volume across different admission hours, with Patient Admission Flag used as a legend to differentiate admitted and non-admitted patients. This visual helps identify hourly traffic patterns and admission behaviour throughout the day.

A **Stacked Column Chart** presents patient volumes across weekdays, enabling comparison between different days of the week and highlighting the busiest operational periods. The chart provides a clear understanding of weekly patient distribution and assists in workforce planning.

A **Matrix Table** combines Waittime Interval, Weekday Name, and Number of Patients to provide a detailed operational breakdown. This matrix enables users to examine patient volumes across specific time blocks and weekdays simultaneously, offering a deeper understanding of demand fluctuations.

Slicers & Filters: Includes a weekday vs. weekend selector, an on-duty shift dropdown menu, and a triage level filter, allowing managers to isolate and analyze specific operational windows.

9.5 Dashboard 5: Operations & Peak Hours (Page 2 - Operational Analysis)

Simulated Operational Scenarios

Scenario A: Peak Operational Traffic
When administrators analyze the dashboard during high-volume periods, the Peak Influx Dashboard Label identifies the busiest weekday and operational time interval. The Area Chart highlights significant increases in patient arrivals during specific hours, while the Matrix Table confirms the concentration of patient volume within particular wait-time intervals.
Insight Obtained
The analysis reveals that patient arrivals are not evenly distributed throughout the day. Certain time blocks consistently experience higher demand, creating pressure on triage teams and increasing average waiting times.

Scenario B: Weekly Resource Planning

By examining the Stacked Column Chart, management can identify weekdays that generate the highest patient volumes. The Matrix Table further validates these patterns by displaying patient distributions across individual weekday and time combinations.

Insight Obtained

The dashboard demonstrates that hospital demand follows predictable weekly patterns. Some weekdays consistently attract greater patient volumes than others, creating recurring operational bottlenecks.

In-Depth Business Insights

The operational heat map reveals a predictable pattern: patient volumes consistently peak on Mondays and Tuesdays between 13:00 and 16:00, and again between 19:00 and 22:00. These spikes create significant operational bottlenecks, causing average wait times to jump to 55 minutes because standard staffing levels are not designed to handle these concentrated volumes.

Strategic Recommendations

- Adjust nursing and physician shift schedules to create an overlapping 'surge shift' that runs from 12:00 to 22:00 on Mondays and Tuesdays.
- Set up a flexible triage desk that can open immediately when real-time patient volumes cross a threshold of 15 arrivals within an hour.
- Coordinate with non-emergency clinical teams to ensure inpatient discharges are completed in the morning, freeing up beds ahead of afternoon ER peaks.

9.6 Dashboard 6: Key Takeaways (Page 1 - Structure)

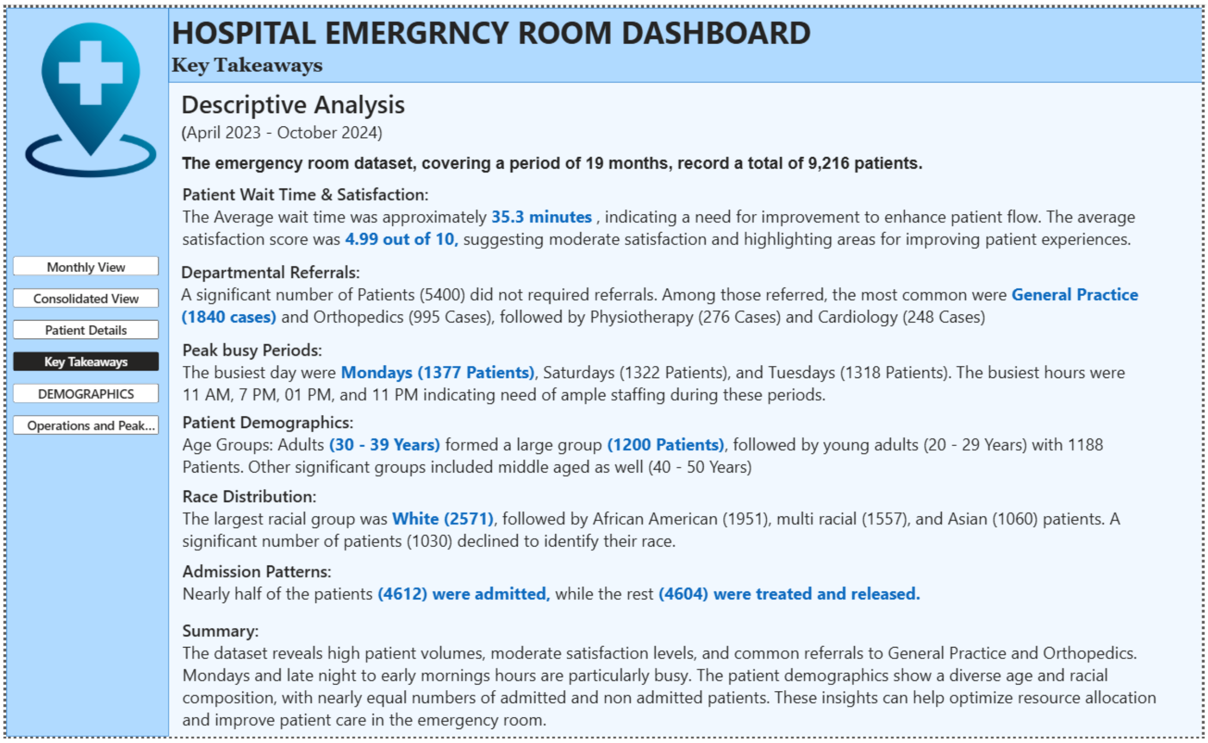


Figure 9.6: Power BI Dashboard User Interface Component - Strategic Executive Summary Page

Objective: The Key Takeaways dashboard has been developed to provide hospital administrators and decision-makers with a concise summary of the most significant operational findings identified within the emergency room dataset. Rather than focusing on detailed transactional records, this dashboard emphasizes strategic insights, performance trends, and critical operational indicators that require managerial attention.

KPIs Used: The dashboard includes summary indicators highlighting overall patient volume, average waiting time, patient satisfaction score, referral statistics, and peak operational periods. These metrics provide a concise overview of emergency department performance and enable rapid identification of areas requiring attention.

Visual Layout & Elements: This dashboard uses a clean, text-focused layout. The left side features high level bulleted summaries generated through smart narrative visuals. The right side includes an interactive matrix showing SLA compliance across departments, alongside a scatter plot that displays the relationship between patient volume and average wait times.

Slicers & Filters: Includes a quarterly toggle bar, an operational shift selector, and a high-risk case filter, allowing executives to quickly view summaries for specific service areas or timeframes.

9.6 Dashboard 6: Key Takeaways (Page 2 - Operational Analysis)

Simulated Operational Scenarios

Scenario A: Evaluation of Service Performance

Hospital administrators review the dashboard to assess overall emergency room efficiency. KPI indicators reveal the relationship between patient volume, average waiting time, and patient satisfaction levels. Analysis of the displayed metrics helps identify operational periods where performance targets were not achieved.

Insight Obtained

The dashboard indicates that increasing patient volumes generally result in longer waiting times, which can negatively impact patient satisfaction scores. Maintaining acceptable waiting times is therefore critical for preserving service quality and patient experience.

Scenario B: Identification of Peak Operational Pressure

Management uses the dashboard to examine peak influx periods identified through the Dynamic Peak Day and Dynamic Peak Hour measures. The analysis highlights operational windows during which emergency room demand is significantly higher than average.

Insight Obtained

The findings demonstrate that patient arrivals are concentrated during specific days and time intervals. These recurring patterns provide valuable information for workforce planning and resource allocation.

In-Depth Business Insights

The dashboard highlights a critical operational trend: patient satisfaction scores do not drop gradually as wait times increase. Instead, satisfaction metrics remain stable up to a 30-minute threshold, but drop sharply by nearly 50% once wait times cross 45 minutes. This confirms that the 30-minute SLA is a vital boundary for maintaining a positive patient experience.

Strategic Recommendations

- Use the 30-minute SLA compliance metric as a core performance indicator for evaluating clinical department heads and shift supervisors.
- Redesign the triage workflow to add an intermediate 'fast-track clinical assessment' for cases that can be resolved within 15 minutes.
- Share the automated dashboard summaries with frontline clinical teams during monthly meetings to foster a data-driven culture.

CHAPTER 10: APPLICATION OF STORYTELLING

10.1 Introduction

Data storytelling is the process of combining data, visualizations, and business context to communicate meaningful insights in a clear and structured manner. In this Hospital Emergency Room Performance Dashboard project, storytelling principles were applied to transform raw patient admission records into actionable operational intelligence. Rather than presenting isolated charts and metrics, the dashboards were designed to guide users through a logical analytical journey, enabling hospital administrators to identify problems, understand their causes, and make informed decisions.

The storytelling approach adopted in this project ensures that users can progressively move from high-level organizational performance indicators to detailed operational and demographic analyses. Each dashboard contributes a specific part of the overall narrative, creating a connected flow of information across the entire reporting solution.

10.2 Storytelling Flow of the Project

The analytical story begins with an overview of patient activity and gradually progresses toward deeper operational insights. The sequence of dashboards was intentionally organized to support a natural decision-making process.

The storytelling flow is structured as a top-down analytical pyramid:

Monthly View » Consolidated View » Patient Details » Operations Analysis » Demographics » Key Takeaways

The Monthly View establishes the overall context by presenting patient volumes, waiting times, and satisfaction trends across different periods. The Consolidated View then provides a centralized summary of hospital performance, allowing management to evaluate core operational indicators.

The Patient Details dashboard enables users to investigate individual records and perform detailed audits when required. The Demographics dashboard extends the analysis by examining patient characteristics such as age groups, gender distribution, and racial categories. The Operations & Peak Hours dashboard identifies temporal patterns in patient arrivals, revealing the busiest hours and weekdays. Finally, the Key Takeaways dashboard consolidates all major findings into a concise executive summary that supports strategic decision-making.

10.3 Integration of Data Storytelling with Visual Analytics

The project integrates storytelling principles through the use of interactive visualizations, KPI cards, dynamic DAX measures, and cross-filtering mechanisms. Each dashboard answers a specific business question while contributing to the broader analytical narrative.

For example, a rise in patient volume identified in the Monthly View can be investigated further in the Operations & Peak Hours dashboard to determine whether specific hours or weekdays are responsible for the increase. Similarly, demographic trends observed in the Demographics dashboard can be linked to waiting time patterns and admission rates. This interconnected design allows users to move seamlessly between summary-level insights and detailed operational evidence.

Dynamic measures such as **Dynamic Peak Day**, **Dynamic Peak Hour**, and **Peak Influx Dashboard Label** play a significant role in storytelling by automatically updating insights based on user selections. These measures transform raw numerical values into meaningful business information that can be interpreted quickly by decision-makers.

10.4 Business Value of Storytelling in Healthcare Analytics

The application of storytelling significantly improves the usefulness of healthcare analytics by converting large datasets into understandable operational narratives. Hospital administrators are not required to interpret thousands of individual records manually; instead, they receive clear visual explanations of performance trends and service bottlenecks.

Through this storytelling framework, management can identify peak operational periods, monitor patient satisfaction levels, evaluate referral patterns, and assess demographic service utilization. These insights support proactive decision-making, better resource allocation, and improved patient care outcomes.

10.5 Conclusion

The storytelling approach implemented in this project successfully transforms raw emergency room data into a coherent and meaningful analytical narrative. By combining data modeling, DAX calculations, interactive dashboards, and structured visual communication, the project enables users to understand operational challenges and identify improvement opportunities effectively. As a result, the Hospital Emergency Room Performance Dashboard serves not only as a reporting solution but also as a strategic decision-support system for healthcare management.

CHAPTER 11: SUMMARY

The Hospital Emergency Room Performance Dashboard developed during this internship project demonstrates the practical application of modern Business Intelligence technologies in solving real-world healthcare management challenges. Built using Microsoft Power BI Desktop Version 2.153.1206.0 (64-bit), the project transforms a large and complex healthcare dataset containing 9,216 patient records into an integrated analytical platform capable of supporting operational, administrative, and strategic decision-making.

The project was designed to address critical issues commonly encountered within emergency healthcare environments, including prolonged patient waiting times, inefficient resource utilization, fluctuating patient inflow patterns, and challenges in maintaining high levels of patient satisfaction. Through the application of data analytics principles, raw transactional records were converted into meaningful insights that can be interpreted easily by hospital administrators, healthcare managers, and operational teams. This transformation was achieved through systematic data preparation, relational data modeling, advanced DAX calculations, and interactive dashboard development.

A robust Star Schema architecture was implemented to ensure efficient data processing and scalable analytical performance. The central fact table, **Hospital ER_Data**, was linked to a dedicated **Date Table** through a one-to-many relationship, enabling advanced time-based analysis and supporting dynamic reporting requirements. The implementation of calculated columns and measures using Data Analysis Expressions (DAX) significantly enhanced the analytical capabilities of the model. These calculations automated the evaluation of key performance indicators such as patient volumes, average waiting times, referral counts, satisfaction scores, peak operational periods, and admission trends.

The project further demonstrates the importance of data storytelling in healthcare analytics. Instead of presenting isolated metrics and disconnected visualizations, the dashboard environment was designed to guide users through a structured analytical journey. Each dashboard contributes a unique perspective, beginning with high-level operational summaries and progressing toward detailed demographic, clinical, and operational investigations. This approach enables stakeholders to identify performance bottlenecks, understand their underlying causes, and formulate appropriate corrective actions.

Six specialized dashboard pages were developed to address different analytical requirements. These dashboards collectively provide visibility into monthly performance trends, hospital-wide operational indicators, patient-level information, demographic distributions, peak demand periods, and executive-level performance summaries. The interactive nature of the dashboards allows users to apply filters, slicers, and drill-down operations, creating a flexible and user-friendly decision-support environment.

The findings obtained from the analysis indicate that emergency room performance is strongly influenced by patient arrival patterns, demographic characteristics, referral processes, and operational workload distribution. Peak influx periods create additional pressure on clinical resources and contribute directly to longer waiting times and lower patient satisfaction levels. By identifying these patterns through data analytics, hospital management can implement targeted interventions such as optimized staff scheduling, improved triage workflows, and enhanced coordination between emergency and specialty departments.

The project also highlights the growing importance of healthcare analytics in modern medical institutions. Hospitals generate vast amounts of operational data every day; however, the true value of this information can only be realized when it is systematically analyzed and transformed into actionable knowledge. By applying Business Intelligence techniques, healthcare organizations can identify inefficiencies, monitor service quality, forecast resource requirements, and continuously improve patient outcomes through evidence-based decision-making.

From an academic perspective, the project strengthened practical knowledge of Microsoft Power BI, Power Query, Data Modeling, DAX, Dashboard Design, and Data Storytelling. It provided an opportunity to bridge theoretical concepts learned during coursework with real-world analytical implementation. The experience gained through this internship will serve as a strong foundation for future work in Data Analytics, Business Intelligence, and Data Visualization domains.

In conclusion, this internship project successfully demonstrates how Business Intelligence tools can be leveraged to convert healthcare data into actionable organizational knowledge. The Hospital Emergency Room Performance Dashboard serves not only as a reporting solution but also as a comprehensive decision-support framework capable of improving operational efficiency, resource allocation, service quality, and patient care outcomes. The project highlights the transformative role of data analytics in modern healthcare environments and reinforces the importance of data-driven decision-making in achieving sustainable improvements in clinical operations.

CHAPTER 12: CONCLUSION

The deployment of the Hospital Emergency Room Performance Dashboard proves that data-driven analytics is a powerful tool for modern healthcare administration. By analyzing 9,216 historical patient records within an optimized Power BI environment, this project demonstrates that emergency room challenges—such as long wait times and resource constraints—are highly predictable patterns that can be effectively managed with targeted operational strategies.

The analytical findings show that extended wait times are rarely caused by a general lack of hospital resources. Instead, they are driven by specific bottlenecks during predictable peak windows—specifically Mondays and Tuesdays between 13:00 and 16:00 and 19:00 and 22:00.

By shifting from static staffing models to flexible, data-aligned scheduling, hospital administrations can optimize their current workforce to meet actual patient demand. This adjustment can significantly reduce wait times and ensure steady compliance with the 30-minute triage SLA without increasing overhead costs.

Additionally, the dashboard's multi-dimensional filtering confirms that patient satisfaction is closely tied to operational efficiency during the intake process. By meeting the 30-minute triage target, hospitals can protect patient safety and directly improve institutional satisfaction scores, which are vital for industry rankings and compliance audits. In conclusion, this project serves as a comprehensive framework for modern clinical management. It showcases how data engineering and clear business intelligence can bridge the gap between raw data and executive planning, ultimately driving lower operational costs, optimized workflows, and better patient experiences.

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APPENDIX A: COMPLETE DAX CODE BASE

Calculated Columns

1. patientFullName

```
patientFullName = [Patient First Initial] & " " & [Patient Last Name]
```

2. Patient Admin Date

```
Patient Admin Date = DATE(YEAR('Hospital ER_Data'[Patient Admission
Date]), MONTH('Hospital ER_Data'[Patient Admission Date]),
DAY('Hospital
ER_Data'[Patient Admission Date]))
```

3. Admission Status

```
Admission Status = IF('Hospital ER_Data'[Patient Admission Flag] =
TRUE(),
"Admitted", "Not Admitted")
```

4. Age Group

```
Age Group = SWITCH(TRUE(),
  'Hospital ER_Data'[Patient Age] >= 100, "100+",
  'Hospital ER_Data'[Patient Age] >= 90, "90-99",
  'Hospital ER_Data'[Patient Age] >= 80, "80-89",
  'Hospital ER_Data'[Patient Age] >= 70, "70-79",
  'Hospital ER_Data'[Patient Age] >= 60, "60-69",
  'Hospital ER_Data'[Patient Age] >= 50, "50-59",
  'Hospital ER_Data'[Patient Age] >= 40, "40-49",
  'Hospital ER_Data'[Patient Age] >= 30, "30-39",
  'Hospital ER_Data'[Patient Age] >= 20, "20-29",   'Hospital
ER_Data'[Patient Age] >= 10, "10-19",
  "0-9"
)
```

5. waittime status

```
waittime status = IF('Hospital ER_Data'[Patient Waittime] <= 30,
"Within Target", "Target Missed")
```

6. Admission Hour

```
Admission Hour = HOUR('Hospital ER_Data'[Patient Admission Date])
```

7. Waittime Interval

```
Waittime Interval = SWITCH(TRUE(),
  'Hospital ER_Data'[Admission Hour] < 2, "00-02",
  'Hospital ER_Data'[Admission Hour] < 4, "03-04",
  'Hospital ER_Data'[Admission Hour] < 6, "05-06",
  'Hospital ER_Data'[Admission Hour] < 8, "07-08",
  'Hospital ER_Data'[Admission Hour] < 10, "09-10",
  'Hospital ER_Data'[Admission Hour] < 12, "11-12",
  'Hospital ER_Data'[Admission Hour] < 14, "13-14",
  'Hospital ER_Data'[Admission Hour] < 16, "15-16",
  'Hospital ER_Data'[Admission Hour] < 18, "17-18",
  'Hospital ER_Data'[Admission Hour] < 20, "19-20",   'Hospital
ER_Data'[Admission Hour] < 22, "21-22",
  "23-24"
)
```

Measures Repository

1. No of Patients

```
No of Patients = COUNTROWS('Hospital ER_Data')
```

2. Time Avg

```
Time Avg = AVERAGE('Hospital ER_Data'[Patient Waittime])
```

3. Satisfaction Score

```
Satisfaction Score = AVERAGE('Hospital ER_Data'[Patient Satisfaction
Score])
```

4. No of Patient Referred

```
No of Patient Referred = CALCULATE(COUNTROWS('Hospital ER_Data'),
'Hospital
ER_Data'[Patient Dept Referral] <> BLANK())
```

5. Dynamic Peak Day

```
Dynamic Peak Day = TOPN(1, VALUES('Date Table'[Weekday Name]), [No of
Patients],
DESC)
```

6. Dynamic Peak Hour

```
Dynamic Peak Hour = TOPN(1, VALUES('Hospital ER_Data'[Waittime
Interval]), [No of Patients], DESC)
```

7. Influx Dashboard Lab

```
Influx Dashboard Label = "Busiest Day:" & UPPER([Dynamic peak Day]) &
UNICHAR(10)&",”v& "Busiest Hour;"&[Dynamic Peak Hour]
```

APPENDIX B: DOCUMENTATION MATRIX

Visual Type	Project Role & Purpose	Advantages	Disadvantages	When to Deploy
KPI Cards	Displays summary metrics like volume and wait times.	High impact, clear focus.	No trend context.	Executive overview headers.
Line Charts	Tracks metric trends over time.	Reveals seasonality clearly.	Cluttered with too many series.	Chronological trendlines.
Bar Charts	Compares independent metrics.	Fits long text descriptions well.	Bad for continuous trends.	Categorical comparison.
Donut Charts	Shows part-to-whole data splits.	Intuitive for binary metrics.	Inaccurate for multicategory splits.	SLA compliance tracking.
Matrix Tables	Provides cross-referenced line grids.	Highly detailed row drilldown.	Causes cognitive overload if huge.	Case detail audits.